But what *are* LLMs?

Neural Networks

The Scale of Operations

100 trillion

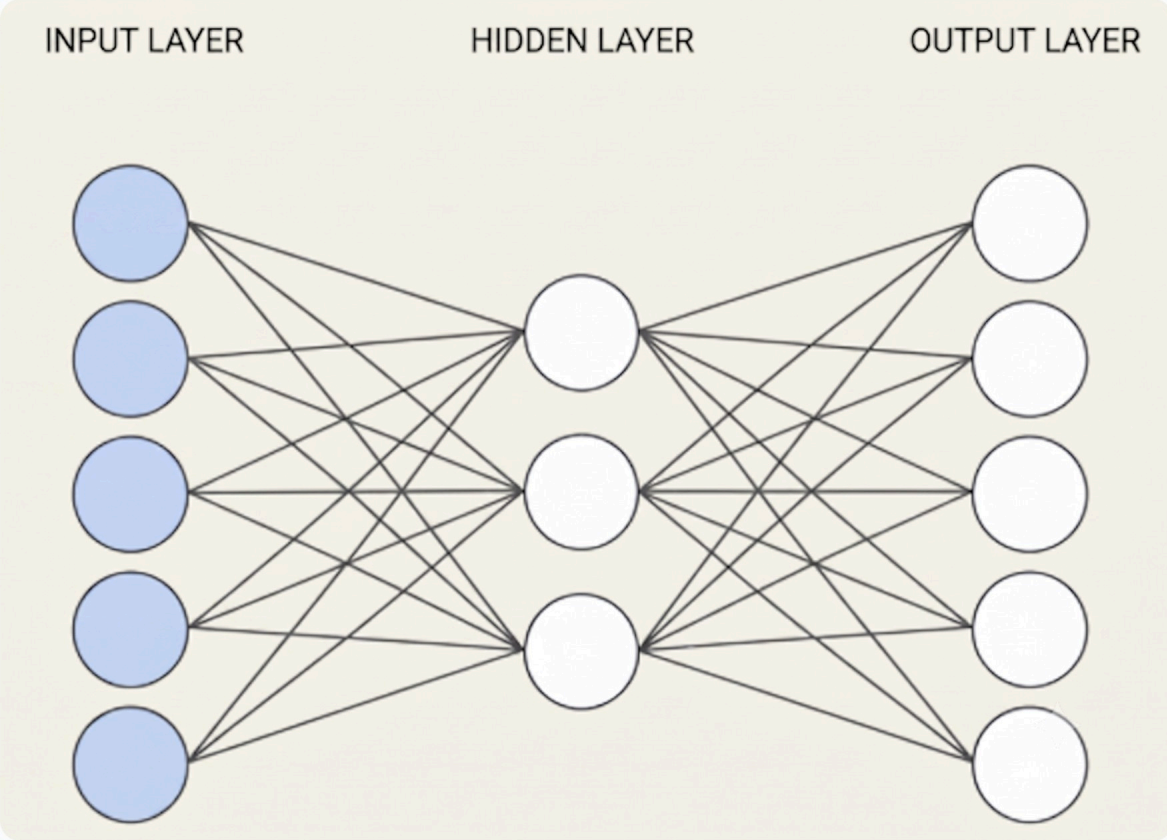
additions and multiplications to answer one question

AI models are **simple math equations**, nothing more.

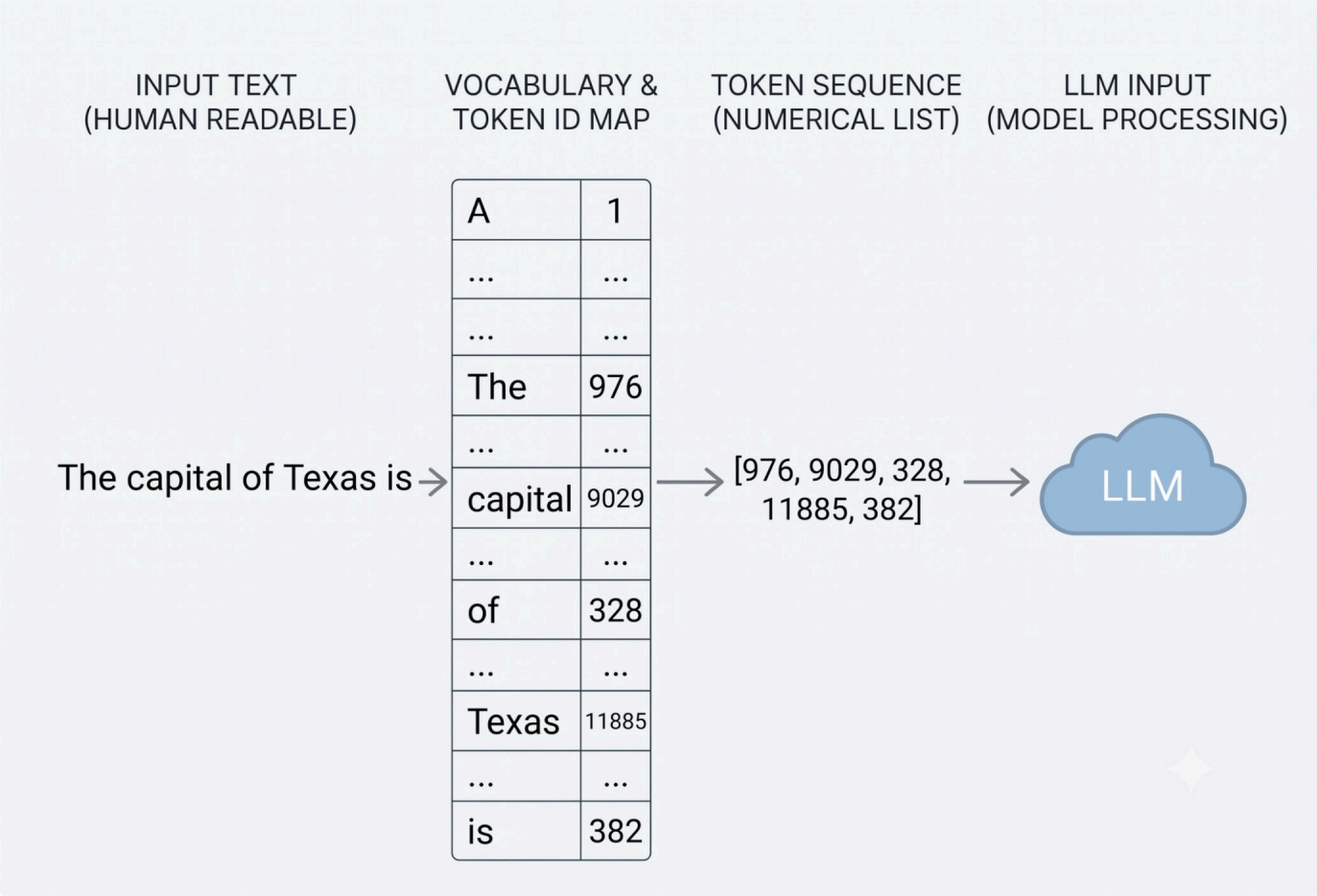
Converting Words to Numbers

A (simplified) **neural network**.

Neural networks only **work with numbers**.



Tokenization



What the Model Actually Sees

Common words

The	capital	of	Texas	is	Austin	.
976	9029	328	11885	382	31628	13

Less common words

M	ito	ch	ond	ria	are	the	powerhouse	of	the	cell	.
44	2955	332	1415	18784	553	290	127147	328	290	4206	13

Try it yourself: platform.openai.com/tokenizer

Trouble Spots: Numbers & Letters

Arithmetic

384	7	×		921	5	=		354	501	05

Spelling

How	many	r	's	are	in	strawberry	?
5299	1991	428	885	553	306	101830	30

Trouble Spots: Scientific Text

DNA

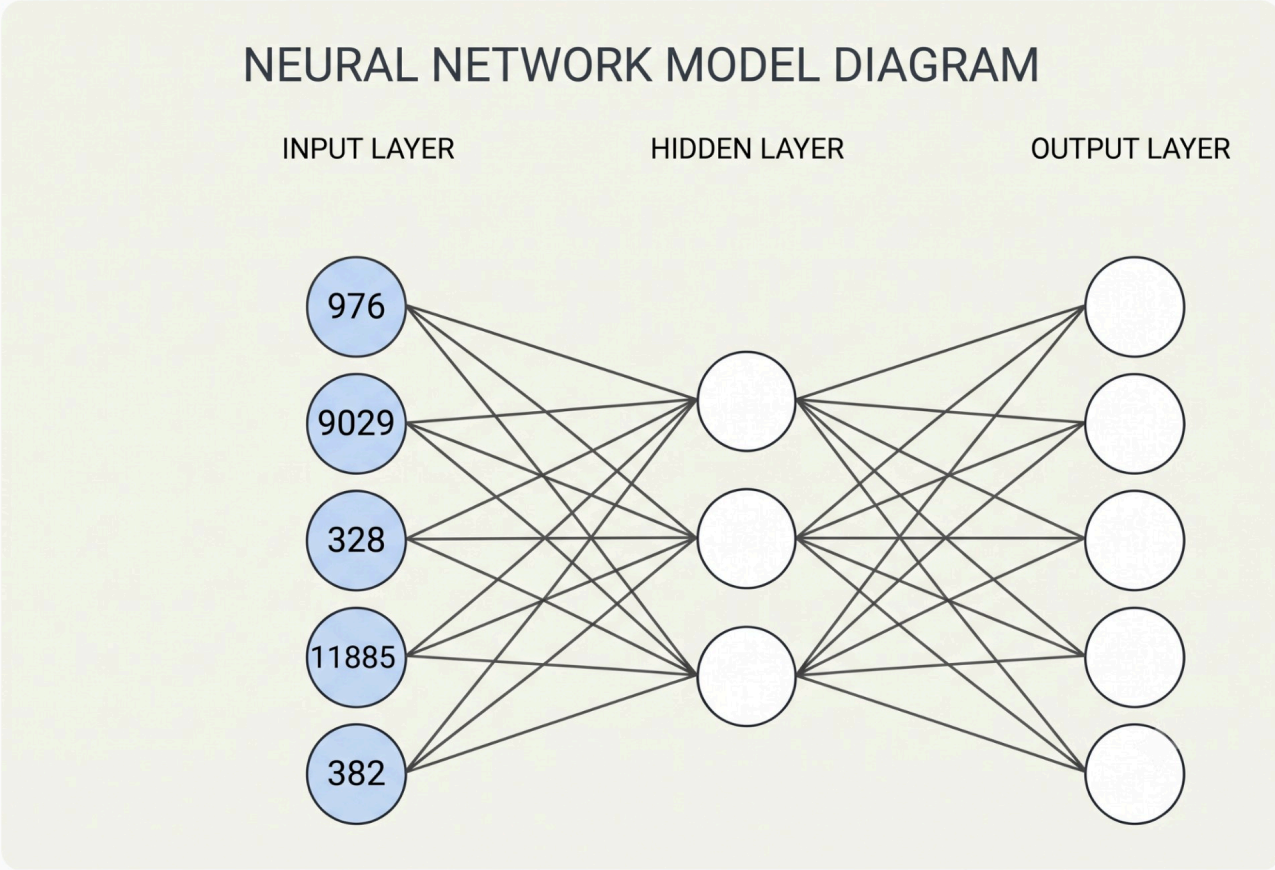
Tokens	AT G CG T AC G TT AG CAT G CA
Codons	ATG CGT ACG TTA GCA TGC A

Chemistry (aspirin)

CC (= O) OC 1 = CC = CC =C 1 C (= O) O

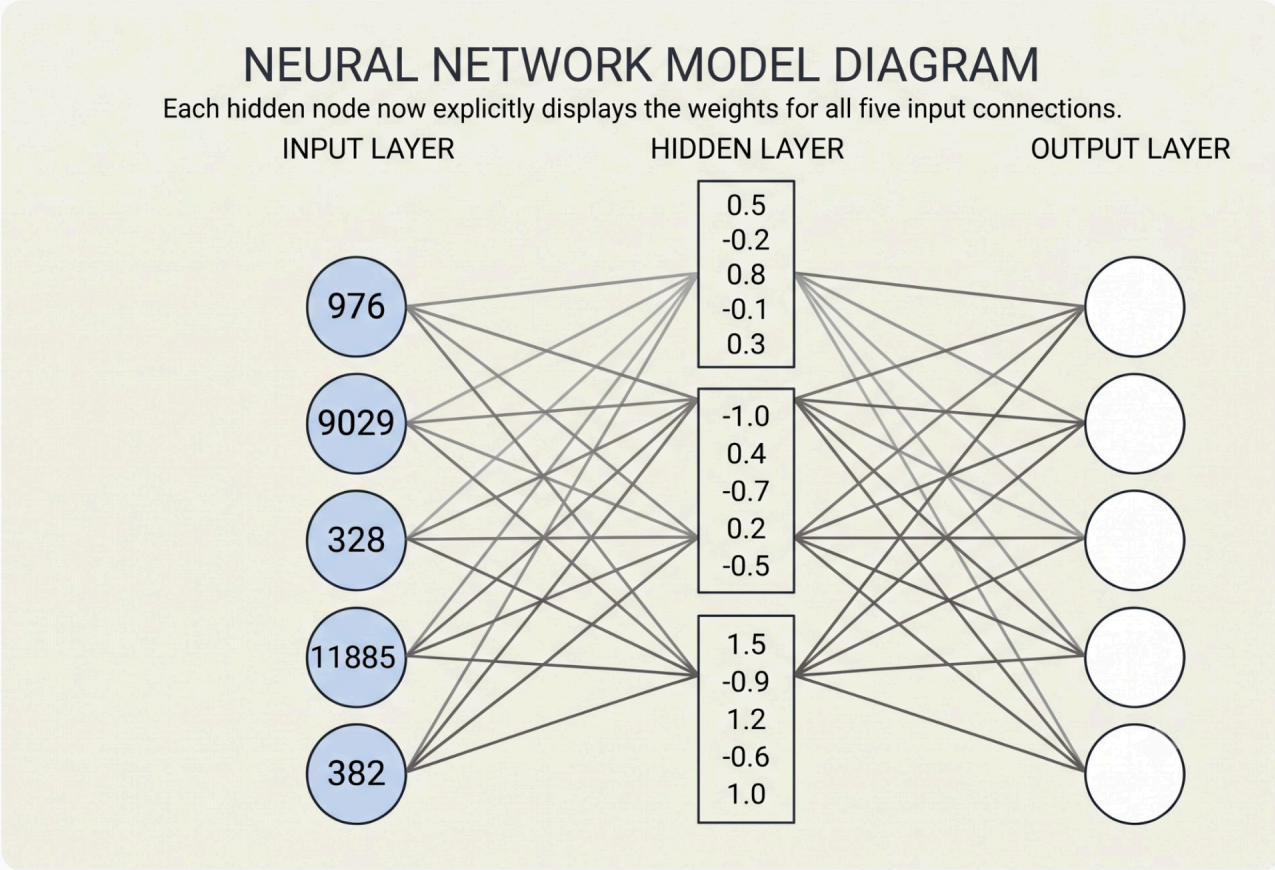
Inside the Network

The token IDs go into the **input nodes**.



Inside the Network

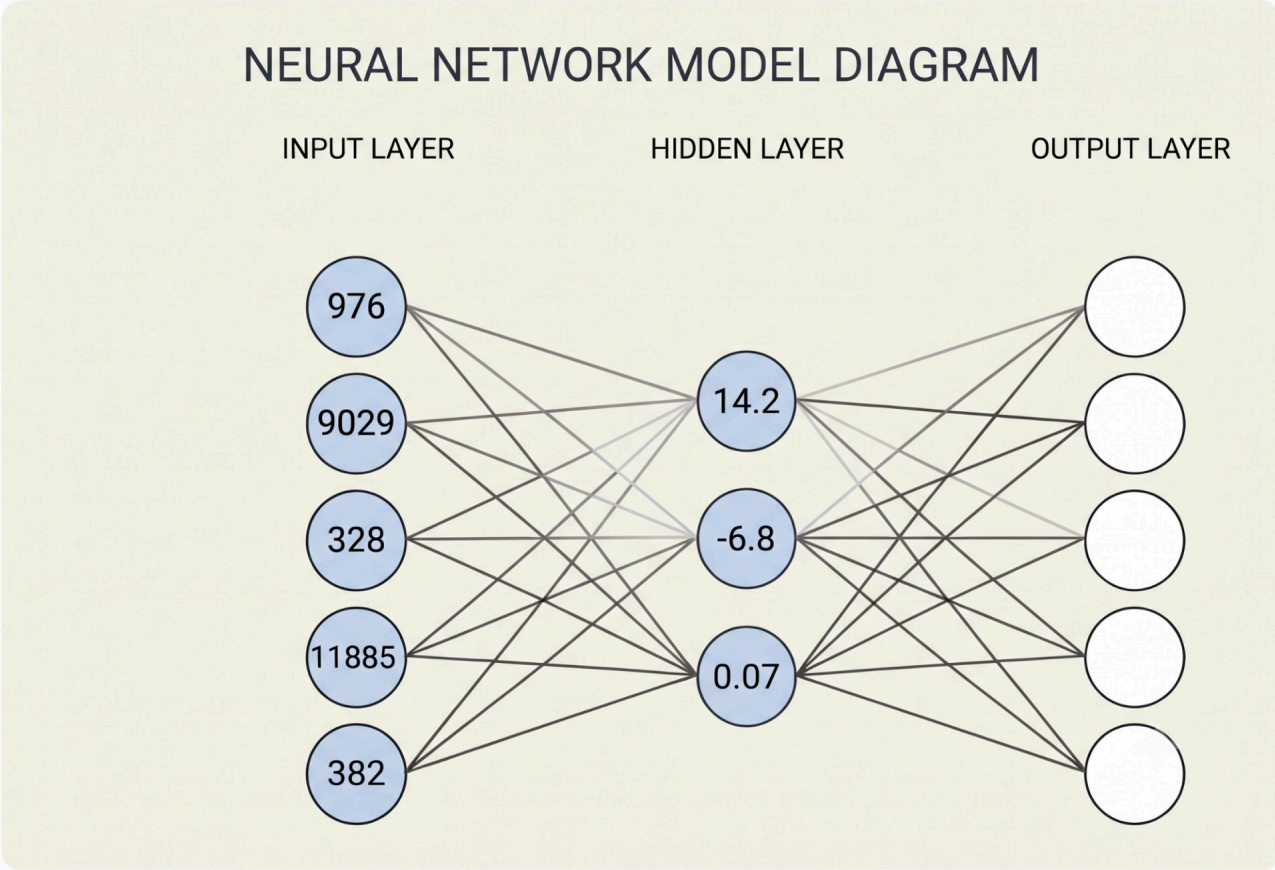
Each hidden node comes with numbers called **parameters** (weights).



Inside the Network

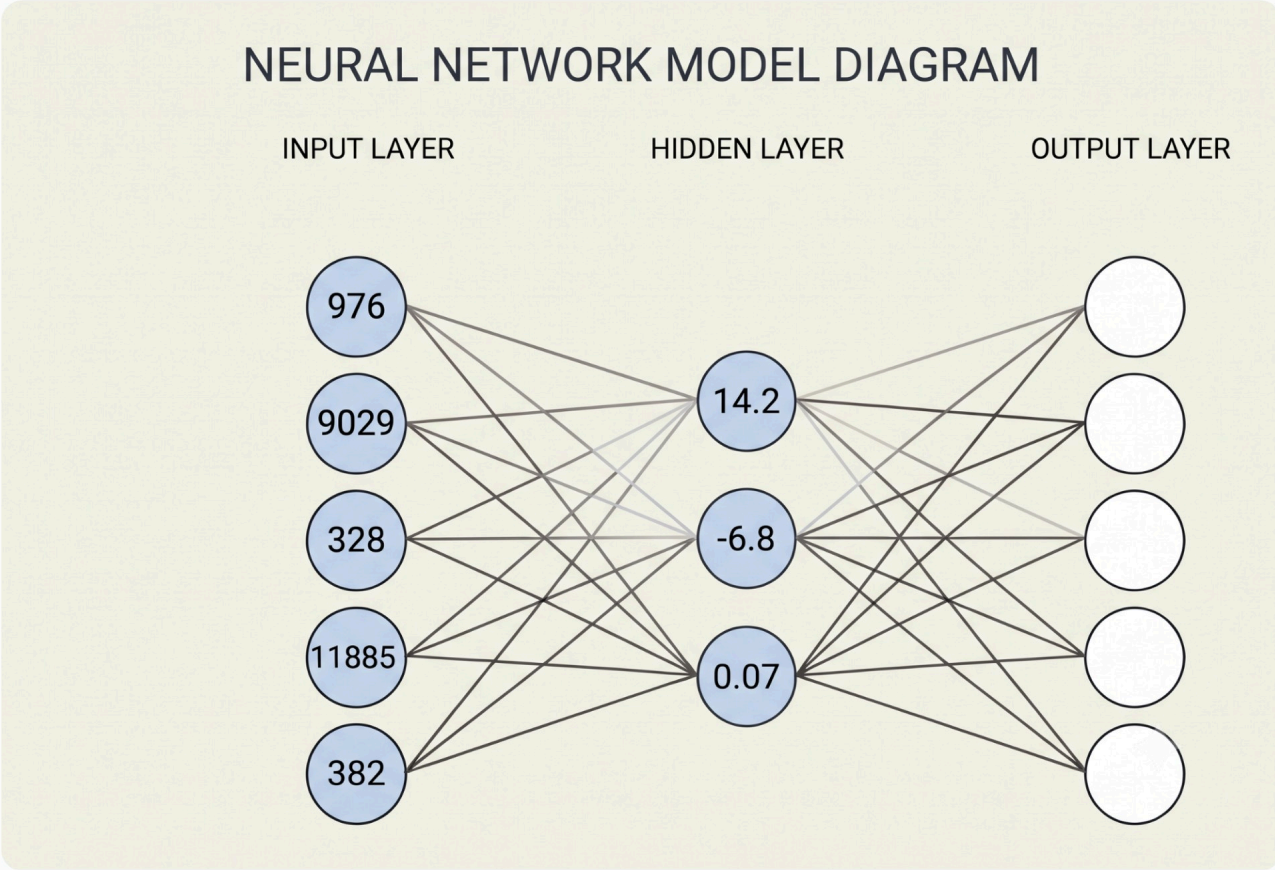
Multiply each input by its weight, then add.

```
  976 × 0.5
+ 9029 × (-0.2)
+ 328 × 0.8
+ …
= 14.2
```



Inside the Network

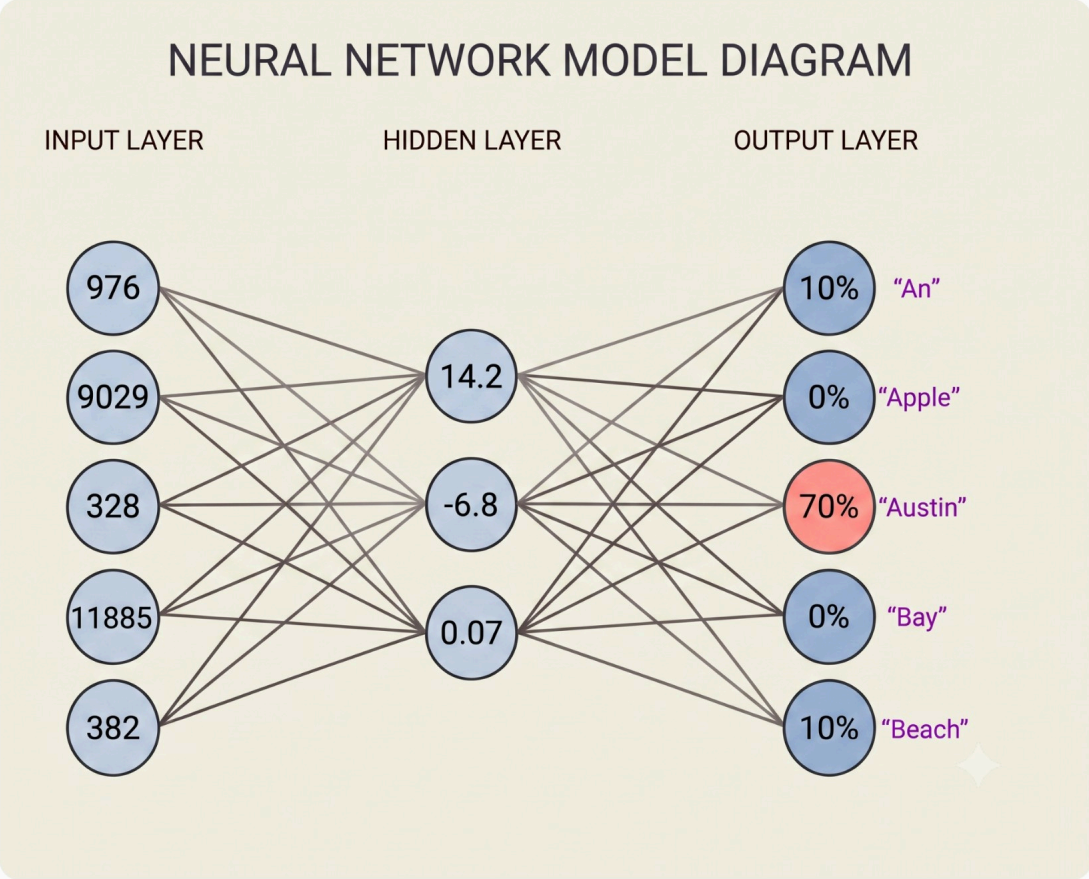
Repeat for every node, layer after layer.



Inside the Network

The outputs are the model's **guesses for the next word**.

Most likely: **"Austin"**



Key Takeaways

Numbers & Math

AI models are just numbers. They convert your input into numbers and perform simple mathematical operations on them.

Prediction

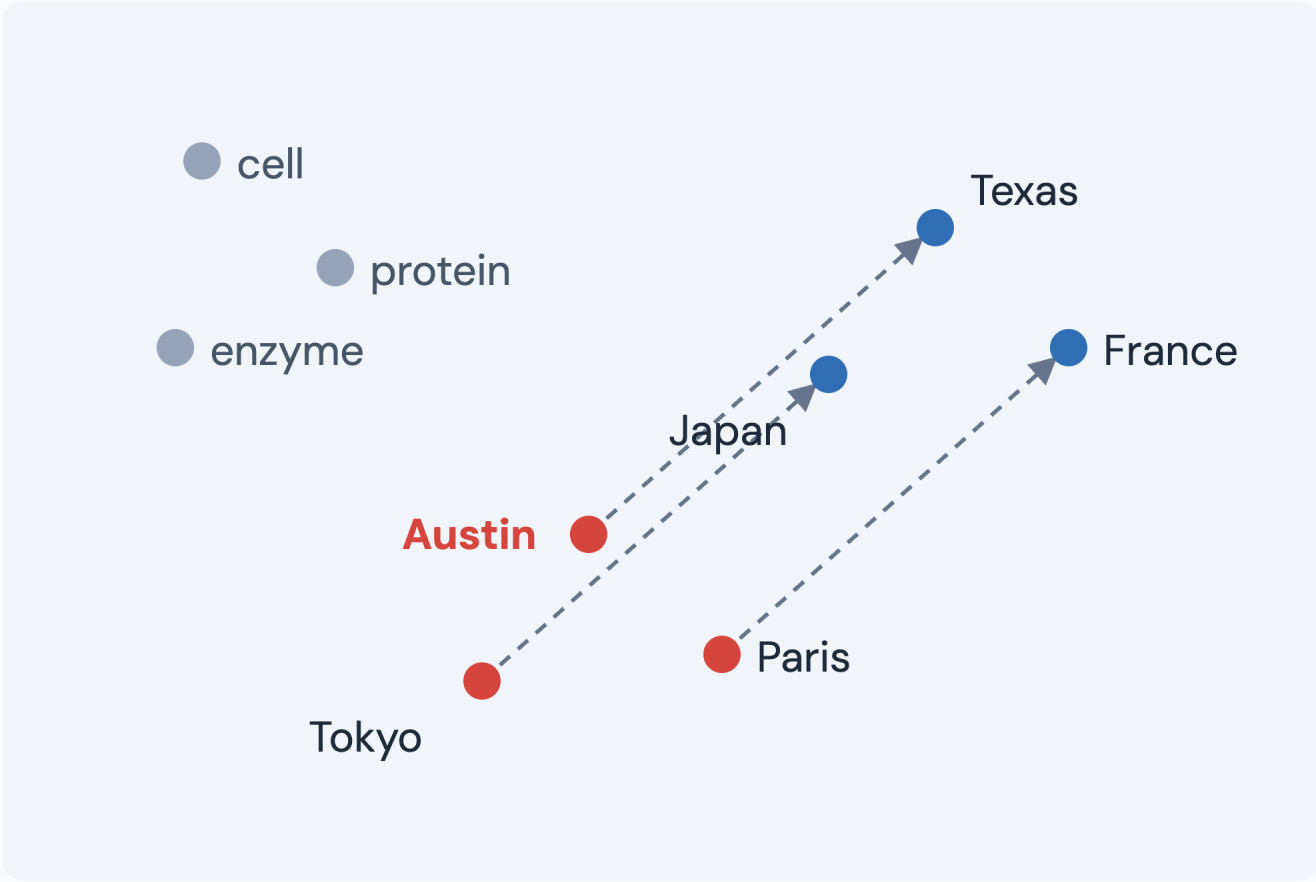
The model predicts which next word is most likely to follow the input words (prompt).

```
Your prompt: “Write me a story about a young boy.”
Model, to itself: What is the most likely first word? Perhaps “Once (upon a time)…”
```

Words as Points in Space

```
Austin →
[0.12, -0.48, 0.91,
 …, 0.07]
```

thousands of numbers per token



Learned by Filling in Blanks

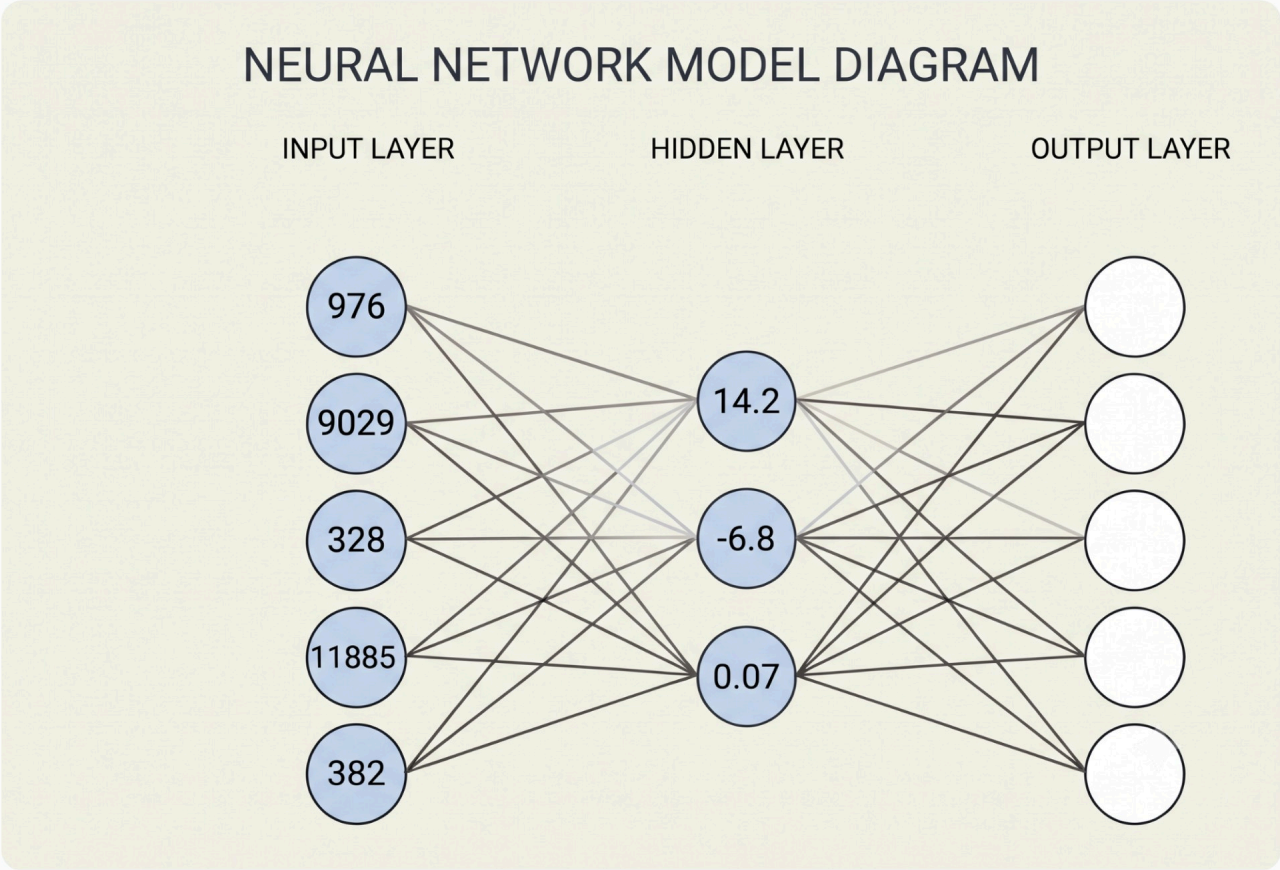
She flew from _____ to Dallas.	Austin, Paris, Tokyo
The _____ folds into its active shape.	protein, enzyme
I added _____ to my coffee.	milk, cream, sugar

Words that fit the same blanks end up **close together**.

Where Do Embeddings Come From?

The numbers in the **hidden layer** are the embedding.

```
[14.2, -6.8, 0.07]
```



But how does the LLM *know* what word should come next?